Cold instruction instrument and resource pilot

Status: independently replayed finite software-count diagnostic; benchmark admission remains incomplete.

The fixed plan contains 174 profiles and 174 attempted executions. It uses six disjoint sizing curves plus the public development controls. No held-out DLP outcome selected this pilot.

Recorded cohort and outcome	Executions
AA / recovered	6
sizing-alpha / recovered	32
sizing-reference / recovered	40
sizing / ambiguous	52
sizing / recovered	44

All 174 raw native/backend totals pass an independent parser. Independent GP checks89 distinct mathematical payloads,41 complete signed witness sets and four generic maps. Six known-answer controls recover2. Native A/A Ir is identical; GP A/A spans101754271–101754276. These are instrument draws, not population variance.

The pilot used523 resource seconds in total. Larger GP-root cases used up to24.3 seconds; group/reference cases used at most1.3 seconds. These clocks only provisioned later limits. Unequal-query pilot arms are not ranked as a performance experiment.

The unit is single-event guest instructions Ir, including actual resident-GP instructions in each exclusive phase. Cold setup, queries, complete decomposition, signs/deduplication, matrix work and source-point verification are charged per target. Input/program loading and result/trace serialization are excluded. No clock, cycle or primitive/GAE conversion is used.

Conditional affine Lean lemmas retain their stated hypotheses. Concrete arithmetic, rank/kernel spans and map fidelity are checked externally; a fully formal concrete bridge is open. Literature and access gaps are in the frozen protocol.

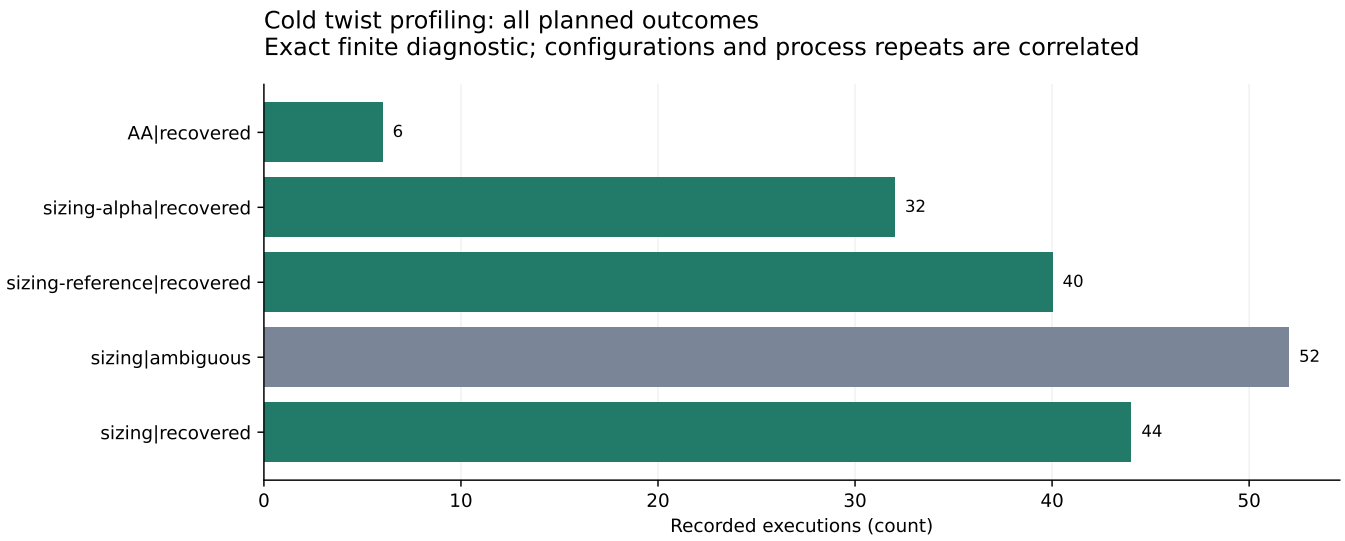


Figure 1: All recorded outcomes

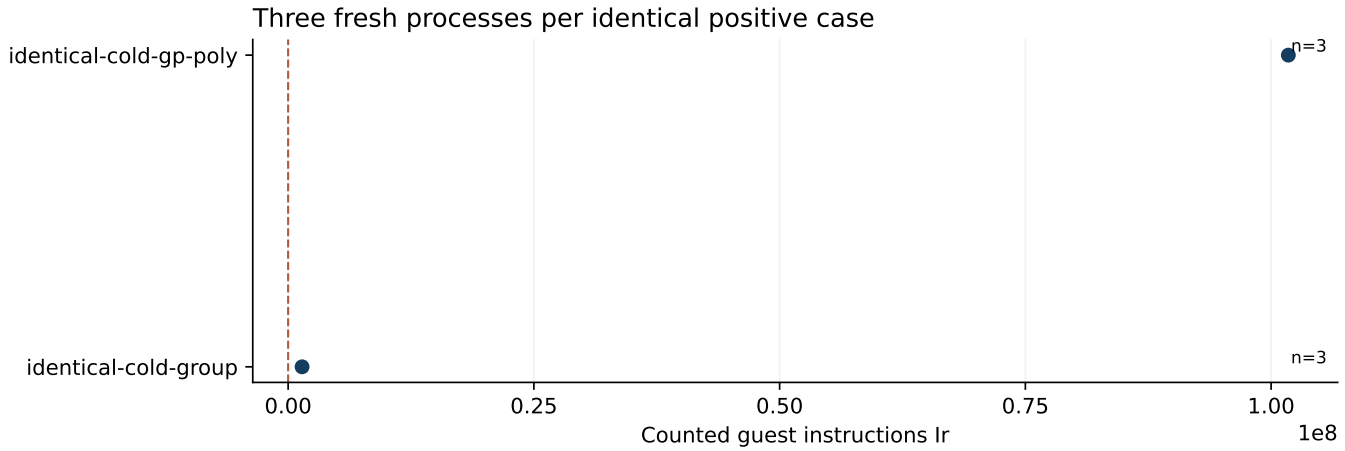


Figure 2: A/A guest-instruction ranges

Sources: frozen protocol, native data manifest, source/byte pins. Independent GP admission.

Reproduction and raw-object manifests are linked from the evidence directory. The user-confirmed store is `setec-astronomy-artifacts`, region us-east-2. No NIST, asymptotic, novelty or calibrated-GAE result is established.